T2T GAPLESS GENOME ASSEMBLY OF THE NEGLECTED CROP *MORINGA STENOPETALA*

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Orphan crops are a largely untapped reservoir of genomic diversity for adapting agriculture to climate change, yet most lack the resources required for modern breeding. *Moringa stenopetala* (Moringaceae), a drought- and heat-tolerant tree native to southern Ethiopia and northern Kenya, is a key food source for vulnerable populations[1,2], but remains semi-domesticated and largely understudied. Here we present a telomere-to-telomere (T2T) assembly of the *M. stenopetala* nuclear genome, generated by combining PacBio Revio HiFi (389.7 × coverage), Hi-C (165.42 × coverage) and Illumina (39.3 × coverage) sequencing. The final assembly spans 267 Mb, with 99.88 % of the sequence anchored to 14 chromosomes (2n = 28), captures 28 of the 28 telomeres and contains 0 gaps (contig N50 = 18.8 Mb; BUSCO completeness = 99.8 %; LAI = 8.78; QV = 44.25). A total of 24,140 protein-coding genes were annotated in the final assembly, 74,89 % of which were supported by RNA-seq expression data, together with 4,052 non-coding RNA genes. Transposable elements account for 9.1% of the genome, primarily composed by LTR retrotransposons (5.4%). All 14 centromeres were fully resolved, spanning 39.6 Mb in total. These are dominated by arrays of a 49-bp satellite repeat (rep49), accounting for 63.5% of the centromeric region, interspersed with TIR transposons (3.9 %), unclassified repeat fragments (3.8 %) and Gypsy LTR retrotransposons (2.9 %). Most centromere genes are transcriptionally silent: only 13.1% of the 1,206 centromeric genes are expressed, compared with 68.2% of genes elsewhere in the genome. We further identified 1,264 nuclear insertions of plastid DNA (NUPTs, 478 kb, 0.18% of the genome) and 2,577 of mitochondrial DNA (NUMTs, 1.40 Mb, 0.52%), together spanning 1.84 Mb (0.69% of the genome). The T2T genome of *M. stenopetala* provides a comprehensive reference for the study of centromere and nuclear organelle DNA evolution. It will also be useful for genetic improvement programs aimed at expanding the cultivation of this resilient plant in arid regions affected by desertification and climate change, notably Almería in southeastern Spain.

Keywords: *Moringa stenopetala*, T2T genome, NUPTs, NUMTs.

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